18.4.1 The eyes - Photoreceptor

The eye is one of the important sensory organs in the human body. It is composed of muscular tissue, connective tissue and neural tissue. It is mainly responsible **for vision, differentiation of color** (the human eye can differentiate approximately 10 – 12 million colors) and **maintaining the biological clock of the human body**. The human eye can be compared to a camera as both functions by gathering, focusing, and transmitting the light through the lens for creating an image of an object.

To understand more about our eye and how our eye functions, we need to look into the structure of the human eye.The human eyes are the most complicated sense organ in the human body, with several parts fixed together forming a spherical structure. Every part of the human eye is mainly responsible for a certain action. The structure of a human eye can be broadly classified into external structure and internal structure.

a. External structure of an Eye

The parts of the eye that are visible externally comprise of the external structure of the eye.

Sclera

It is a tough and thick white sheath that protects the inner parts of the eye. We know it as the **'white of the eye'.**

Conjunctiva

It is a thin transparent membrane that is spread across the sclera. It keeps the eyes moist and clear by secreting small amounts of mucus and tears.

Cornea

It is the transparent layer of membrane that is spread over the pupil and the iris. The main role of the cornea is to refract the light that enters the eyes.

Iris

It is a pigmented layer of tissues that make up the colored portion of the eye. Its primary function is to control the size of the pupil, depending on the amount of light entering it.

Pupil

It is the small opening located at the middle of the iris. It allows light to come in.

b. Internal structure of an Eye

The internal structure of the eye includes the following parts.

Lens

It is a transparent, biconvex, and an adjustable part of an eye, made up of protein. The lens with the help of the cornea refracts light which converges on the retina and creates images on it.

Retina

It is the layer present at the back of the eye where all the images are formed. The retina functions by converting the light rays into impulses and sending the signals to the brain through the optic nerve.

Optic nerve

It is located at the end of the eyes, behind the retina. The optic nerve is mainly responsible for carrying all the nerve impulses from the retina to the human brain.

Aqueous Humour

It is a watery fluid that is present in the area between the lens and the cornea. It is

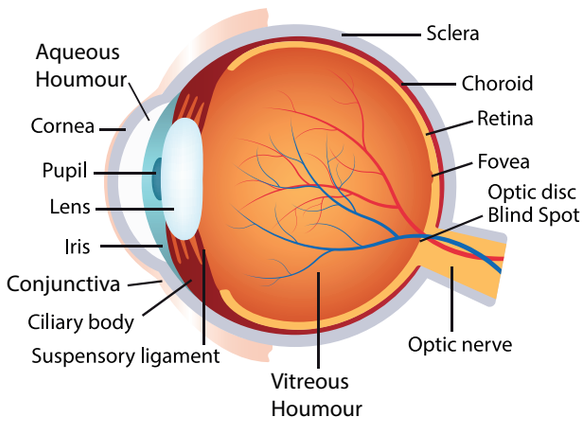


Figure 18.6 Structure of Human Eye

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